APPLICATIONS OF ULTRAFAST PROCESSES

EXAMPLES OF QUESTIONS FROM PREVIOUS EXAMS

Example 1.
In the framework of the third-order optical nonlinear polarization, in the semi-classical approach for light-matter interaction and using the rotating-wave approximation, describe the non-collinear pump-probe experiment for a two-level system. In particular, using the perturbative approach, describe which are the main steps to compute the third-order polarization along the probe direction starting from the equations describing the temporal evolution of the $\rho_D^{(n)}$ and $\tilde{\rho}_{21}^{(n)}$ terms of the density matrix as a function of their (*n-1*) counterparts of the power series.

Example 2.
Discuss the difference between homogeneous and inhomogeneous broadening in a two-level system. In the framework of the third-order optical nonlinear polarization, employing the perturbative approach and the density-matrix formalism in the semi-classical approach for light-matter interaction, describe the two-pulse photon-echo experiment and how it can be used to extract the homogeneous dephasing time T_2 for the two extreme cases of purely homogeneous broadening and of very large inhomogeneous broadening. <u>Note</u>: you can start from the assumption that the pump and probe fields are Dirac-delta functions centred at t=0 and t=τ: $A_{pu} = A_1\delta(t)$ and $A_{pr} = A_2\delta(t-\tau)$ and the third-order coherence term of the density matrix propagating in the $2\vec{k}_{pu} - \vec{k}_{pr}$ direction in the rotating frame can be written as:
$\tilde{\rho}_{21,2pu-pr}{}^{(3)} = -i\left(\frac{\mu}{2\hbar}\right)^3 A_1 {A_2}^2 e^{-\frac{t}{T_2}} e^{-i\Delta\omega t} e^{2i\Delta\omega\tau} He(\tau) He(t-\tau)$

Example 3.
<u>Framework</u>: within the semiclassical approach, consider a weak electromagnetic field $E(t) = \frac{1}{2}\{A(t)e^{i\omega_0 t} + c.c.\}$ impinging onto a two-level system with energies $\mathcal{E}_1$ and $\mathcal{E}_2$. Using perturbation theory (note: <u>not</u> the perturbative approach for the density matrix!), consider that the energy levels and eigenstates of the perturbed Hamiltonian can be expressed in terms of those found solving the time-independent Schrödinger equation: $|\psi\rangle = c_1(t)|u_1(\vec{r})\rangle e^{-i\frac{\mathcal{E}_1}{\hbar}t} + c_2(t)|u_2(\vec{r})\rangle e^{-i\frac{\mathcal{E}_2}{\hbar}t}$.
<u>Questions:</u>

a) Using the rotating-wave approximation and the dipole approximation, solve the time-dependent Schrödinger equation for such a case, thus extracting the expressions that describe the temporal evolution of the two coefficients $\frac{dc_1}{dt}$ and $\frac{dc_2}{dt}$.
b) Now consider an extremely short laser pulse at t=0, which can be approximated by a Dirac-delta function $A(t) = A_0\delta(t)$, impinging onto a two-level system initially prepared in the ground state. Compute the polarization (also said "polarization density" or "electric polarization") and describe its evolution as a function of time after the pulse. Why is this result not possible in reality? Explain the need for the density matrix formalism to solve this issue.

Example 4.
Consider optical waves at frequencies ω_1, ω_2, ω_3 and ω_4, such that ω_1-ω_2+ω_3=ω_4, impinging in a medium with a third-order non-linear response. In the framework of the four-wave mixing process and using the slowly varying envelope approximation, discuss the SRS (stimulated Raman scattering)

process. Compute the $\frac{\Delta I}{I}$ signal (in the case of SRG or SRL), discussing its dependence on the intensities of the beams, on the propagation length and on $\chi^{(3)}$.

Example 5.
Introduce the concept of pure and mixed states in quantum mechanics and the density matrix operator, discussing the physical interpretation and role of the diagonal and off-diagonal elements. Introduce the longitudinal and transverse relaxation processes, explaining why they cannot be introduced using the wavefunction picture.

Example 6.
In the context of non-degenerate broadband pump-probe experiments:

- Describe the general optical setup and the role of each optical element.
- Introduce the two-dimensional ΔT/T signal as a function of wavelength and delay and discuss the PA (photo-induced absorption), PB (photo-bleaching) and SE (stimulated emission) signals.
- Explain the need for lock-in amplifiers and their working principle.
- Finally, discuss the temporal resolution of a pump-probe experiment.

Example 7.
Consider optical waves at frequencies ω_1, ω_2, ω_3 and ω_4, such that $\omega_1-\omega_2+\omega_3=\omega_4$, impinging in a medium with a third-order non-linear response $\chi^{(3)}$. In the framework of the four-wave mixing process and using the slowly varying envelope approximation, discuss the CARS (coherent anti-Stokes Raman scattering) process. Calculate the intensity of the detected signal, discussing its dependence on the pump and Stokes intensities, on the propagation length and on $\chi^{(3)}$. Briefly discuss the role of the non-resonant background (NRB) and the dependence on the sample concentration.

Example 8.
Consider the non-collinear degenerate pump-probe experiment in a linear continuous-time system with causal impulse response function $h(t)$. Starting from the Lambert-Beer law, in the small signal limit, calculate the signals detected in transmission configuration by a slow detector along the probe direction. You can limit the discussion to the pump-probe terms and skip those related to the coherent coupling artefacts.